Musculoskeletal control for functional electrical stimulation-assisted rowing

Achille Roulot[1,2] and Ana C. C. de Sousa[2]

1 Seatech-School of engineering, Université de Toulon, France. 2 Institut Quimic de Sarria - Universitat Ramon Llull (IQS-URL), Spain, ana.desousa@iqs.url.edu

Introduction

Individuals with spinal cord injury (SCI) often have a sedentary lifestyle, increasing the risk of cardiovascular, metabolic and bone-density complications [1]. Functional electrical stimulation (FES) activates paralysed muscles using electrical currents and, in rowing, combines stimulated lower-limb muscle with voluntary upper-limb motion, providing a whole-body cardiovascular exercise [2, 3]. However, reliable FES rowing requires muscle activation to be synchronised with the stroke phase. This work develops and evaluates an automatic controller in an OpenSim musculoskeletal simulation that links upper-limb kinematics to lower-limb stimulation, enabling control strategies to be tested in silico before patient application.

Methods

Voluntary upper-limb motion was reproduced using forward dynamics and recorded motion-capture trajectories from a rower (Fig. 1). A closed-loop computed-torque controller (CTC) adjusted shoulder, elbow, forearm and trunk joint torques to track the reference motion. Elbow angle represented stroke position, while angular velocity indicated movement direction; together these signals drove a three-state finite-state machine (FSM) commanding lower-limb extension, no stimulation or flexion. Elbow signals were smoothed using a 0.5-s window. State transitions used angle thresholds of 20° and 130° and a ±10°/s velocity dead-band to prevent rapid switching. Extension activated bilateral vastus intermedius and gluteus maximus, while flexion activated sartorius/gracilis, biceps femoris short head, and psoas. The CTC and FSM operated at 300 and 50 Hz, respectively.

Results

The CTC reproduced the recorded elbow trajectory (right elbow RMSE 3.37°, Fig 2). Elbow angle and angular velocity identified the main stroke phases and drove the three-state FSM, alternating lower-limb activation between extension, no stimulation and flexion (Fig 2). The controller maintained the appropriate state during the imposed 2-s pauses and resumed switching when motion continued. A sliding-seat translation limit constrained leg excursion and prevented knee hyperextension, enabling a complete simulated rowing cycle.

Conclusions

These results demonstrate an in silico automatic controller for FES-assisted rowing that links voluntary upper-limb motion to lower-limb stimulation. Elbow angle and angular velocity provided simple signals for stroke-phase identification and stimulation switching. The current controller provides a proof of concept; its switching thresholds and stimulation settings should be adapted to individual biomechanics and muscle response before application in people with SCI. Future work will transfer the strategy to hardware and validate the approach with people with SCI.

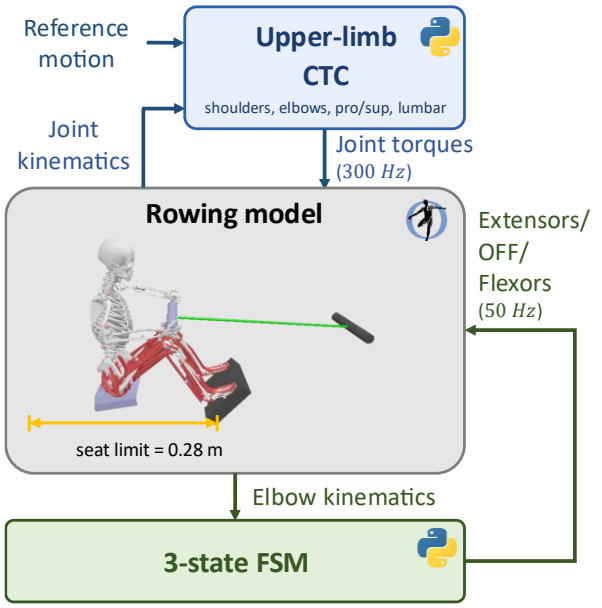


Figure 1: Overview of the in silico control framework for FES-assisted rowing. A closed-loop CTC uses reference and simulated joint kinematics to reproduce voluntary upper-limb motion in OpenSim. Elbow angle and angular velocity drive a three-state finite-state machine that activates the lower-limb extensors, flexors or neither.

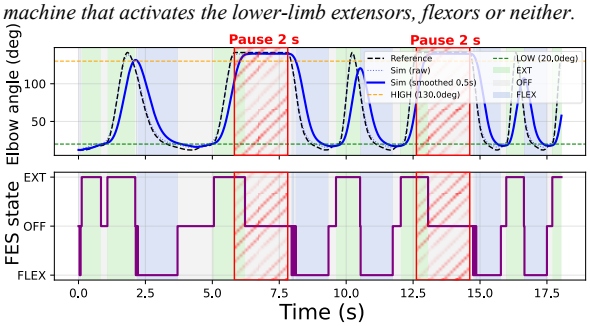


Figure 2: Representative controller response showing elbow-angle tracking and the resulting three-state FES command. Background colours indicate extension, no stimulation and flexion, while hatched regions denote 2-s pauses.

Acknowledgements

The authors thank Isabelle Ormerod at the University of Bristol (UK) for providing the motion-capture reference data.

References

1. Itodo et al, Eur J Epidemiol, 37:335-365, 2022.
2. Wheeler et al, Arch Phys Med Rehabil, 83:1093-1099, 2002.
3. Ye et al, Arch Phys Med Rehabil, 102:1595-1605, 2021.